

Benchmark Comparisons for Active STAR-RIS Assisted UAV Networks With NOMA

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This supplementary note provides additional benchmark comparisons for the proposed active STAR-RIS assisted UAV network with NOMA. In the main paper, the proposed scheme has been compared with the passive STAR-RIS assisted case. To further demonstrate the performance advantage and practical significance of the proposed model, we supplement comparative experiments with the following benchmark schemes: (1) UAV-NOMA network without STAR-RIS assistance; (2) RIS-assisted UAV network with NOMA; (3) RSMA-based active STAR-RIS assisted UAV network.

For fairness, the main simulation parameters are kept the same as those in the second experiment group of the main paper (Section III.B), including the UAV transmit power, user mobility setting, fading parameters, imperfect CSI parameter, residual SIC parameter and SINR thresholds. For the RIS-assisted UAV network with NOMA benchmark, the number of reflecting elements is set to be the same as the number of STAR-RIS components. For the UAV-NOMA benchmark without STAR-RIS assistance, no intelligent surface is deployed and the UAV directly serves the terrestrial users. For the RSMA-based benchmark, the same active STAR-RIS assisted UAV framework is adopted, while the multiple access technique is replaced by RSMA.

I. COMPARISON WITH UAV-NOMA NETWORK WITHOUT STAR-RIS ASSISTANCE

As we know, the UAV-NOMA scheme without STAR-RIS can be regarded as a baseline case, where the UAV directly serves terrestrial users without any intelligent surface assistance. The parameters setting is the same as the second experiment group (Section III.B). Here we only show the basic derivation about downlink transmission of UAV-NOMA network without STAR-RIS device or relay.

Since UAV directly transmits the superimposed signal to $\mathcal{U}_\kappa$, the distance of “UAV to user” is denoted as $d_{u\kappa}$. Hence, taking $\mathcal{U}_2$ as the example, its SINR for SIC process and decoding its desired signal can be respectively expressed as

$$\gamma_s = \frac{(1-a)Pd_{u\kappa}^{-o_3}|\mathbf{h}_2|^2}{aPd_{u\kappa}^{-o_3}|\mathbf{h}_2|^2 + P\sigma_e^2 + \sigma_0^2} \quad (1)$$

$$\gamma_2 = \frac{aPd_{u\kappa}^{-o_3}|\mathbf{h}_2|^2}{\beta_{RI}(1-a)Pd_{u\kappa}^{-o_3}|\mathbf{h}_2|^2 + (1+\beta_{RI})aP\sigma_e^2 + \sigma_0^2} \quad (2)$$

where d_{u2} denotes the distance between UAV and $\mathcal{U}_2$ and o_3 represents the corresponding path loss exponent of the direct

UAV-user link. Based on the above formula, we can further derive the downlink metrics of UAV-NOMA network without STAR-RIS device or relay (we do not show the detailed derivation, since it is enough simple).

fig. 1 shows the detailed comparisons between our study and the downlink performance of UAV-NOMA network without STAR-RIS device or relay, noting that the increase of number of STAR-RIS components N does not affect the downlink performance of this benchmark scheme. It is easy to find that the proposed active STAR-RIS assisted UAV-NOMA scheme achieves significantly better performance than the benchmark UAV-NOMA scheme in terms of coverage probability CP , delay outage rate DOR , ergodic capacity EC and energy efficiency EE . The reason is that the active STAR-RIS introduces additional power gain and improves the equivalent channel quality, whereas the UAV-NOMA benchmark does not have such assistance.

It is also worth noting that, when the number of STAR-RIS components N is relatively small, the benchmark UAV-NOMA scheme may exhibit better performance than the considered STAR-RIS assisted UAV-NOMA scheme. This is because for small N , the power gain provided by STAR-RIS is still limited. Meanwhile, both ES and MS protocols lead to a splitting effect at STAR-RIS side. As a result, the effective cascaded channel gain $|\chi_\kappa|^2$ cannot be sufficiently enhanced, and the gain brought by STAR-RIS is unable to fully offset the corresponding performance loss when N is small. By contrast, the UAV-NOMA benchmark relies on direct transmission and does not suffer from this effect. Hence, its performance can be better when N is small. However, as N increases, the power gain brought by STAR-RIS becomes dominant, such that the proposed STAR-RIS assisted UAV-NOMA scheme gradually outperforms the UAV-NOMA benchmark.

With the rapid development of STAR-RIS technique, its hardware and deployment costs would be further reduced. Hence, compared with the UAV-NOMA network without using STAR-RIS or relay, utilizing STAR-RIS assisted UAV-NOMA architecture is more promising.

II. COMPARISON WITH RIS-ASSISTED UAV NETWORK WITH NOMA

As we know, the proposed model in our work benefits from the full-space transmission capability of STAR-RIS, whereas the traditional RIS can only support users in a single side region. Hence, the two schemes are inherently different in system architecture and service coverage. Accordingly, the purpose of this comparison is to provide a meaningful

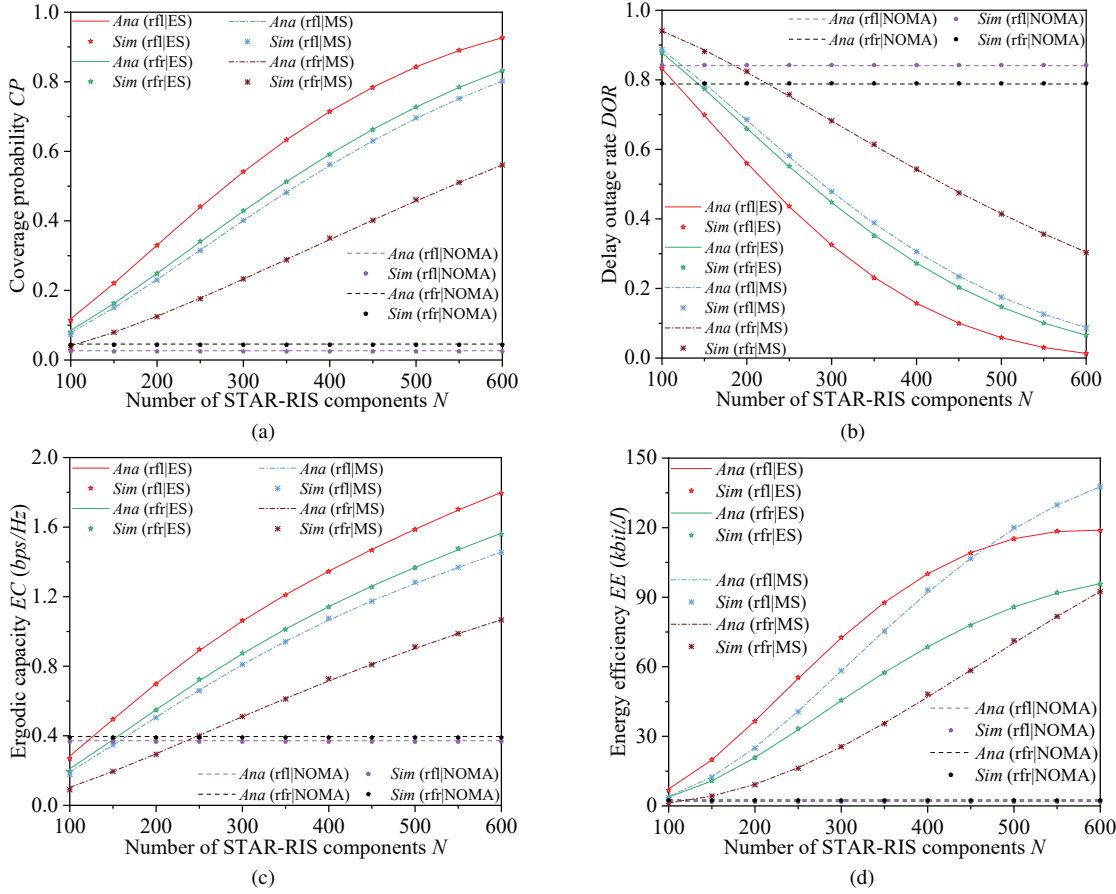


Fig. 1. Our work versus UAV-NOMA network without STAR-RIS assistance

benchmark for illustrating the performance advantage of our proposed model, rather than to pursue a strictly one-to-one system equivalence.

Since the conventional RIS-assisted UAV network with NOMA can be regarded as a simplified special case of our proposed model, the corresponding SINR expressions and related downlink metrics can be directly obtained by simplifying the proposed analytical framework. Specifically, compared with the active STAR-RIS assisted case, the traditional RIS does not involve the thermal noise introduced by active amplification. In addition, the RIS benchmark only supports single-side transmission, and thus it does not have the full-space service capability of STAR-RIS. Under the considered fair comparison setting, the same UAV transmit power, power allocation coefficient, user mobility setting, fading parameters, imperfect CSI parameter, residual SIC parameter, SINR thresholds and number of surface elements are adopted.

As shown in fig. 2, it is easy to find that the proposed active STAR-RIS assisted UAV-NOMA scheme generally achieves better performance than the RIS-assisted UAV network with NOMA benchmark across all metrics. The reason is that the active STAR-RIS introduces additional power gain through active amplification, which improves the equivalent channel quality and the received SINR. Moreover, executing ES protocol yields the best overall performance, while executing MS protocol still outperforms the RIS-assisted UAV network with NOMA benchmark in most cases. It is worth noting that

the comparison of EE is not unique. Although the proposed active STAR-RIS assisted scheme achieves better EE when N is not very large, the RIS-assisted UAV network with NOMA benchmark may exhibit comparable or even better EE when N becomes sufficiently large. This is because the active amplification introduces additional power consumption, while RIS-assisted scheme operates with lower power consumption. In addition, $\mathcal{U}_2$ achieves better performance than $\mathcal{U}_1$ under all considered schemes, which is consistent with the resource allocation setting in our paper.

Also, it should be stressed that the above comparisons still provide meaningful benchmarks for evaluating our proposed model, although the benchmark schemes are not fully equivalent to the proposed active STAR-RIS assisted UAV-NOMA network in terms of system configuration and operating mechanism. For example, the conventional UAV-NOMA scheme does not utilize any intelligent surface, while the RIS-assisted UAV network with NOMA scheme only supports single-side transmission and does not include active amplification. Even so, the above comparative results still clearly demonstrate the superior anti-attenuation capability of active STAR-RIS assisted UAV networks with NOMA. Therefore, investigating active STAR-RIS assisted UAV networks with NOMA is indeed very necessary.

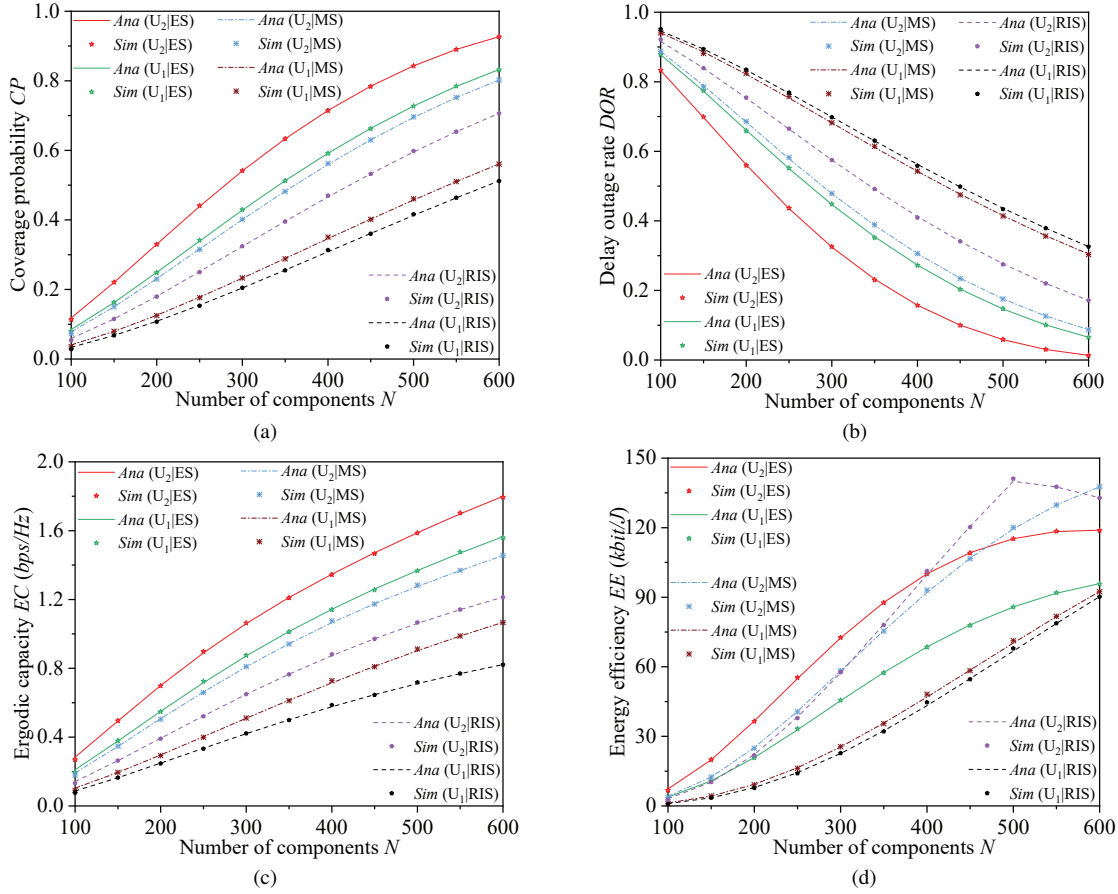


Fig. 2. Our work versus RIS-assisted UAV network with NOMA

III. COMPARISON WITH RSMA-BASED ACTIVE STAR-RIS ASSISTED UAV NETWORK

As we know, RSMA has emerged as another promising multiple-access technique for STAR-RIS assisted wireless communication systems. Comparing STAR-RIS assisted UAV communication systems with NOMA and STAR-RIS assisted UAV communication systems with RSMA is meaningful for illustrating the different influence of multiple access technique on the end-to-end downlink transmission performance in the considered hybrid mobile environment. Therefore, we further supplement the corresponding comparative experiment.

For RSMA assisted communication systems, the superimposed signal transmitted from UAV to STAR-RIS can be rewritten as

$$X = \sqrt{a_c P} x_c + \sqrt{a_2 P} x_2 + \sqrt{a_1 P} x_1 \quad (3)$$

where a_c , a_2 and a_1 are the transmission power allocation coefficients for the common message stream, the private message of reflecting user $\mathcal{U}_2$ and the private message of refracting user $\mathcal{U}_1$, respectively, with $a_c + a_2 + a_1 = 1$. x_c and x_κ ($\kappa \in \{1, 2\}$) are the transmitted common message and private message from UAV, respectively, with $\mathbb{E}[|x_c|^2] = \mathbb{E}[|x_\kappa|^2] = 1$.

In the active STAR-RIS assisted UAV network with RSMA, $\mathcal{U}_\kappa$ first decodes the common message by treating all private message as interference, and then decodes its own private message while treating the other user's private message as

interference. Hence, the SINR expressions can be written as (taking $\mathcal{U}_2$ with the execution of MS protocol as the example)

$$\gamma_{2,c} = \frac{\rho^2 a_c P \Phi_2^2 |\chi_2|^2}{\rho^2 \left((a_2 + a_1) P \Phi_2^2 |\chi_2|^2 + \sigma_1^2 d_{s2}^{-\alpha_2} |G_2|^2 \right) + P \sigma_e^2 + \sigma_0^2} \quad (4)$$

$$\gamma_{2,p} = \frac{\rho^2 a_2 P \Phi_2^2 |\chi_2|^2}{\rho^2 \left(a_1 P \Phi_2^2 |\chi_2|^2 + \sigma_1^2 d_{s2}^{-\alpha_2} |G_2|^2 \right) + P \sigma_e^2 + \sigma_0^2} \quad (5)$$

Using the same parameters setting of the second experiment while setting $a_c = 0.4$ and $a_1 = a_2 = 0.3$, fig. 3 shows the influence of number of active STAR-RIS components N on the downlink performance, as well as the detailed comparisons between our NOMA-based system and its RSMA-based counterpart.

It is easy to find that as the number of active STAR-RIS components increases, the downlink performance of active STAR-RIS assisted UAV network is improved under both RSMA and NOMA. Moreover, the proposed NOMA-based active STAR-RIS assisted UAV network generally achieves better downlink performance than its RSMA-based counterpart in terms of both coverage probability CP and ergodic capacity EC . Similarly, the reflective user attains better downlink

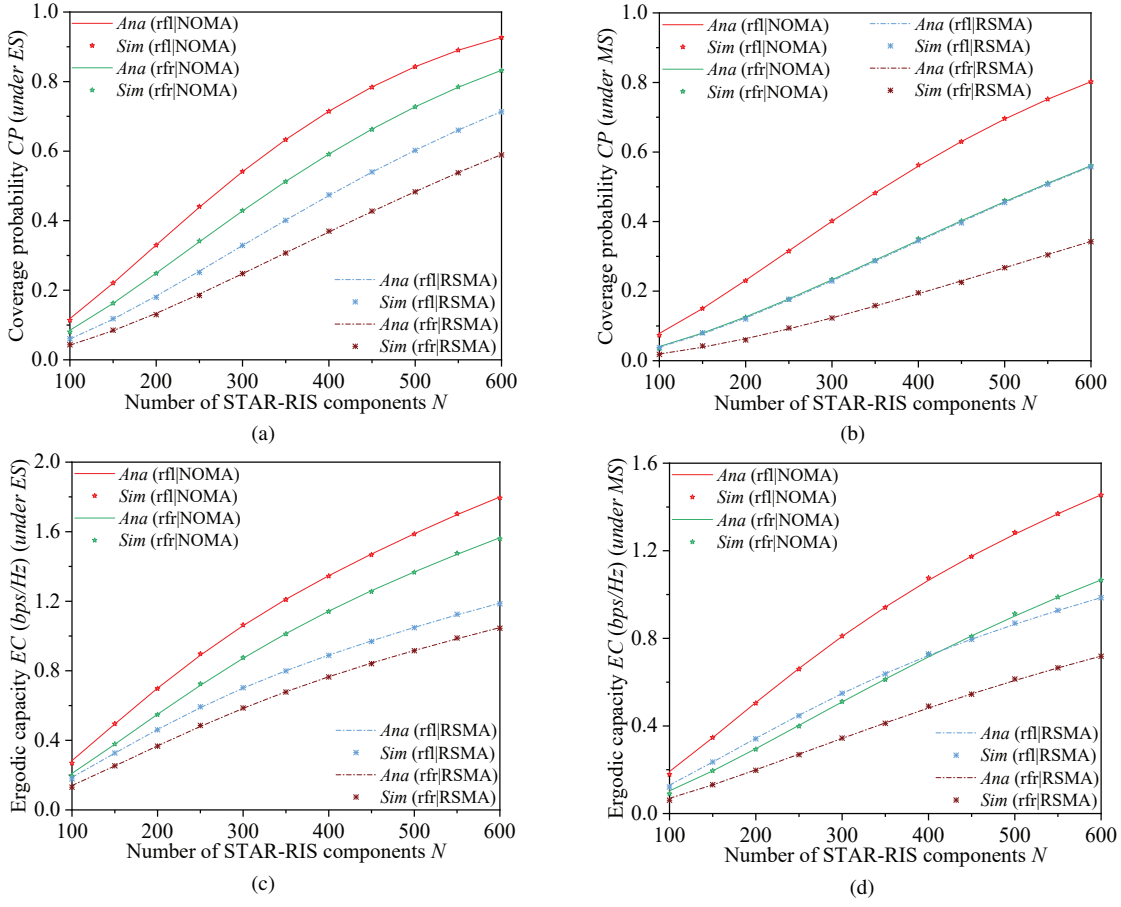


Fig. 3. Our work versus RSMA-based system

performance than refractive user, and executing ES protocol yields superior performance compared with executing MS protocol, which is consistent with the analysis in our article.

It should be noted that a strictly one-to-one parameter equivalence between the NOMA-based system and its RSMA-based counterpart is difficult to guarantee, especially in terms of transmission power allocation coefficients. This is because the RSMA-based system involves an additional common stream, and thus its power allocation structure is different from that of the NOMA-based system. Even so, the above comparison still provides a meaningful benchmark for illustrating the relative performance characteristics of the two multiple-access schemes in the considered hybrid mobile environment.